

MIDAP: A Full-Stack Open NPU Platform for Embedded AI Exploration

University Demonstration at DAC 2026

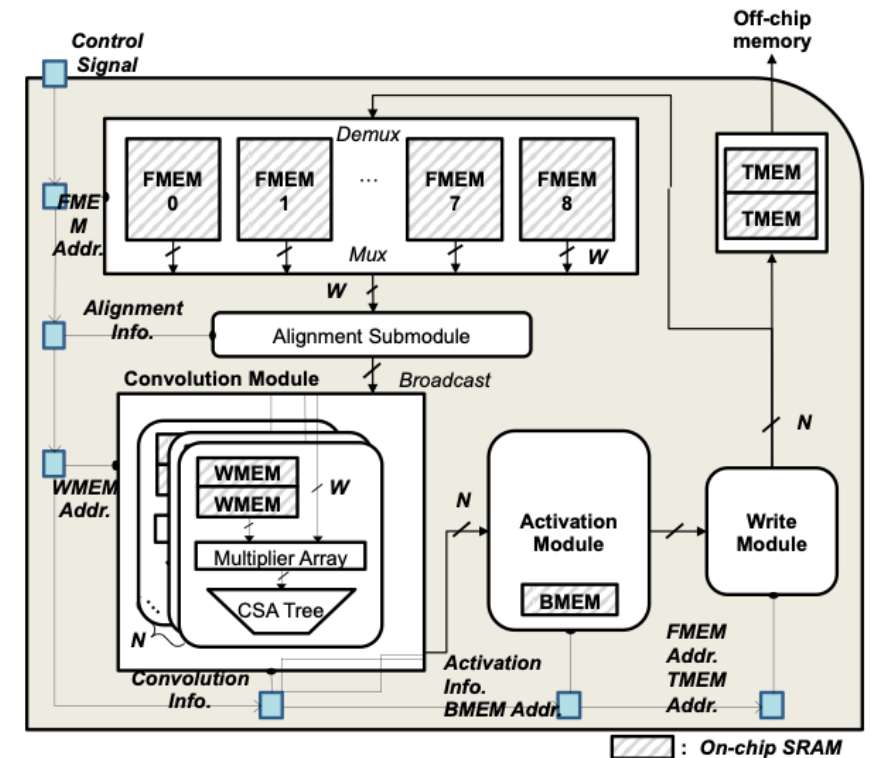
Changjae Yi, Hyunsu Moh, Seongwoo Choi, Joon Choi,
Youngchul Yoon, and Soonhoi Ha
Co-design and Parallel Processing Laboratory
Dept. of Computer Science and Engineering
Seoul National University

Motivation

- Edge AI is becoming local
 - Embedded systems increasingly run AI inference locally
 - Embedded deployment constraints vary widely
- Closed NPU stacks make exploration difficult
 - hard to inspect compiler decisions
 - hard to modify architecture
- MIDAP's role
 - Provides a full HW/SW path for embedded NPUs
 - Connects model, IR, simulator behavior, microcode, and RTL-facing artifacts
- Planned to be released as open source soon

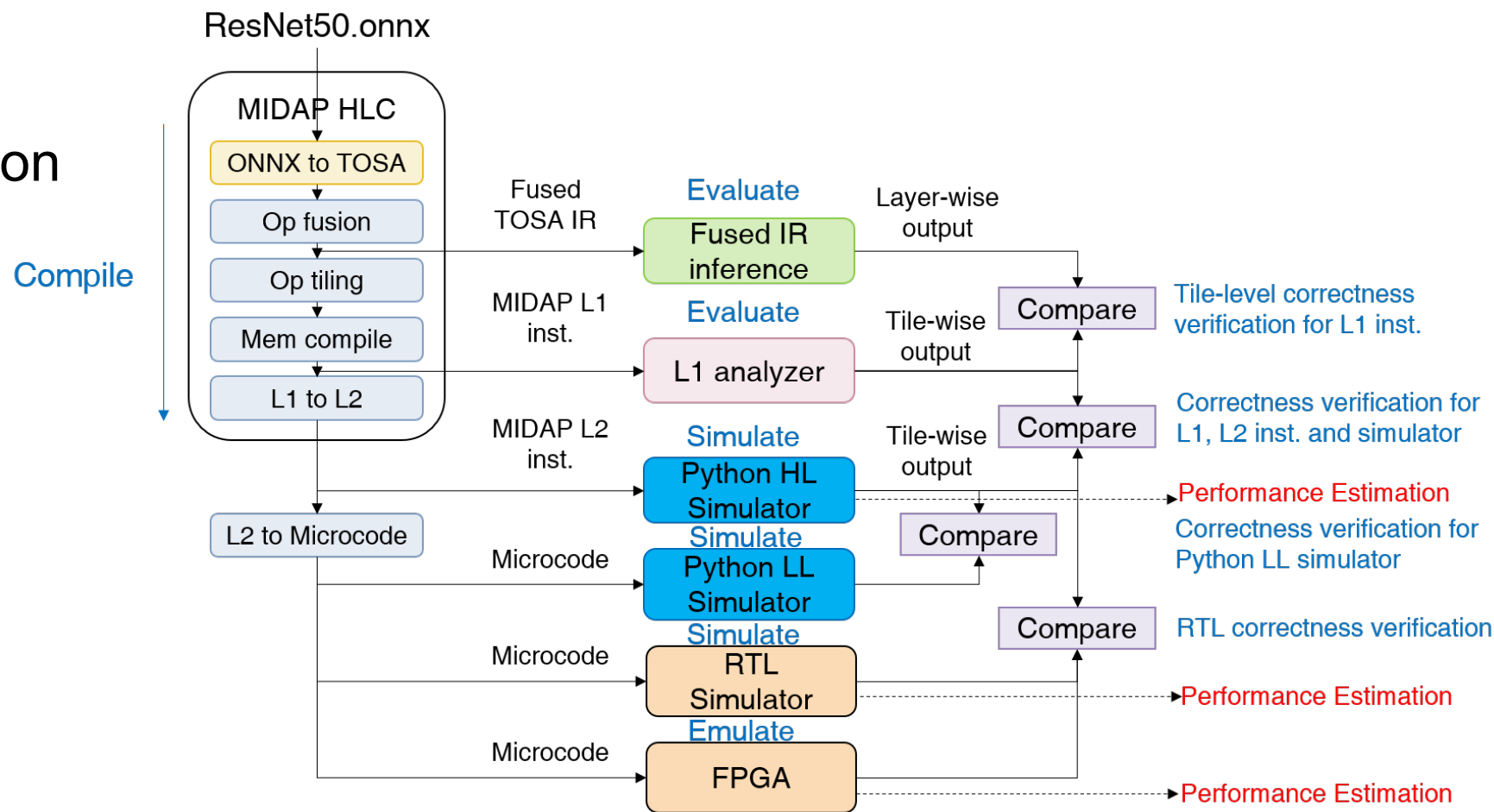
MIDAP NPU

- Memory-In-the-Datapath Architecture Processor
 - Adder-tree based, fully-pipelined
- A high-performance, energy-efficient neural processor for embedded AI
 - High MAC utilization: Resnet50(47.5%), Efficientnet-b2(16.9%), YOLOv4 tiny(74.3%), Deeplabv3+(75.1%)
- Supports a wide range of workloads
 - image classification
 - object detection
 - image segmentation
 - and transformer models like ViT



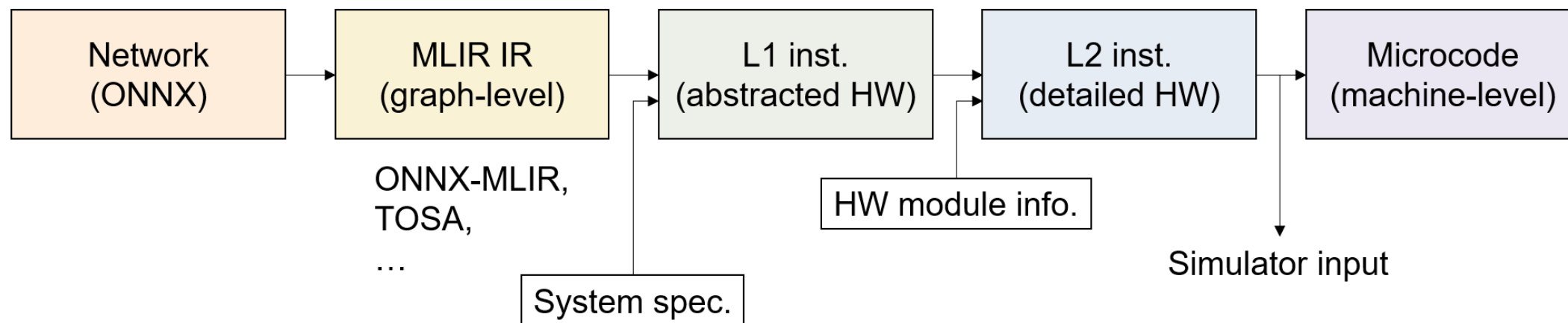
End-to-End Workflow

- **Compiler:** translates AI models into target instructions
- **Simulator:** simulates the workload at high-level, verifies and evaluates the architecture
- **RTL & FPGA:** low-level verification



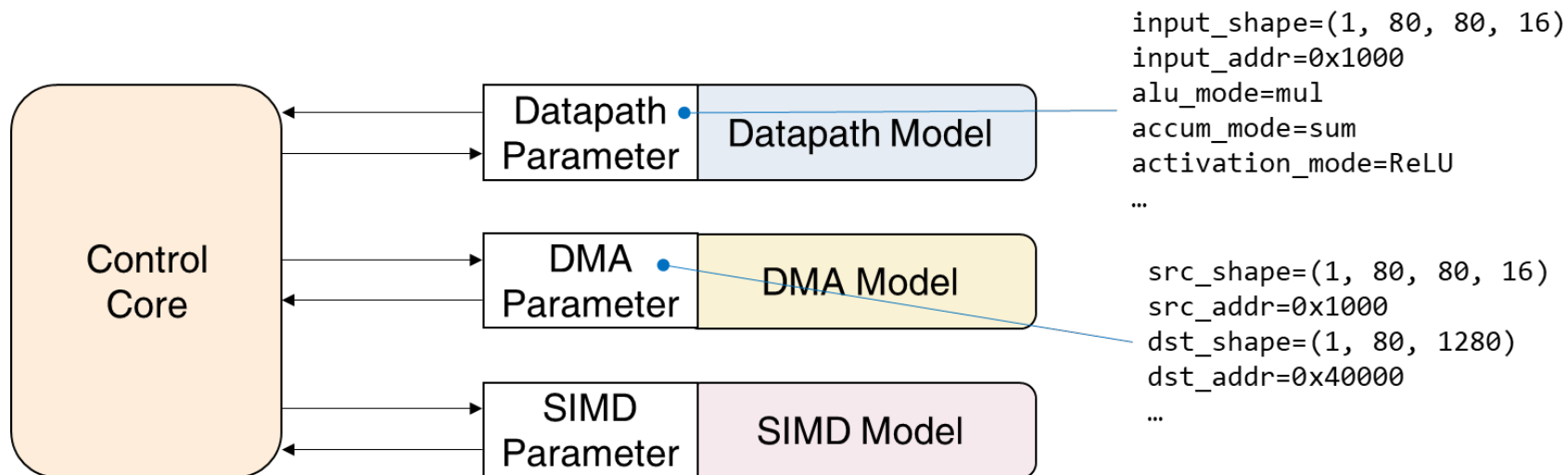
Compiler

- Converts ML models into executable instructions for the NPU HW and/or simulator
- MLIR-based design – friendly to the open-source community
- Two levels of NPU IR (L1/L2)
 - L1: abstracts away low-level controls, focused on tile-level scheduling
 - L2: detailed HW control instructions with exact parameters and values
 - Makes it easy to accommodate HW modifications



Simulator

- Provides a high-level virtual prototype for HW/SW co-evaluation before RTL
- Supports cycle-level accurate performance estimation and functional verification
 - Enables efficient design space exploration (DSE) in early design stages
- Modular and extensible architecture



RTL/FPGA Implementation

- Uses the same workload artifacts across simulation and RTL validation
- Validates datapath, decoder, and control logic at RTL level
- Closes the loop between compiler, simulator, and hardware-facing execution

